Superstore Sales Performance Analysis

Data Analytics Internship Task | Future Interns | Python, pandas, matplotlib

Objective

Help a retail business understand which products, categories and regions drive revenue and profit, how sales change over time, and where to focus to grow faster.

Dataset

Sample Superstore: four years of US orders (Jan 2014 to Dec 2017), 9,994 order lines and 21 columns. No missing values or duplicate rows were found.

Approach

- **Cleaning:** standardised column names, converted dates, checked duplicates, added Year, Month, Profit Margin and Shipping Days.
- **KPIs:** total sales, profit, margin, orders, customers and average order value.
- **Analysis:** trends by month and year, category, sub-category, region, segment, state, discount level and top products.
- **Dashboard:** single-page visual summary with insights and recommendations.

Headline KPIs

Total sales	Total profit	Margin	Orders	Customers	Avg order
$2.30M	**$286K**	**12.47%**	**5,009**	**793**	**$459**

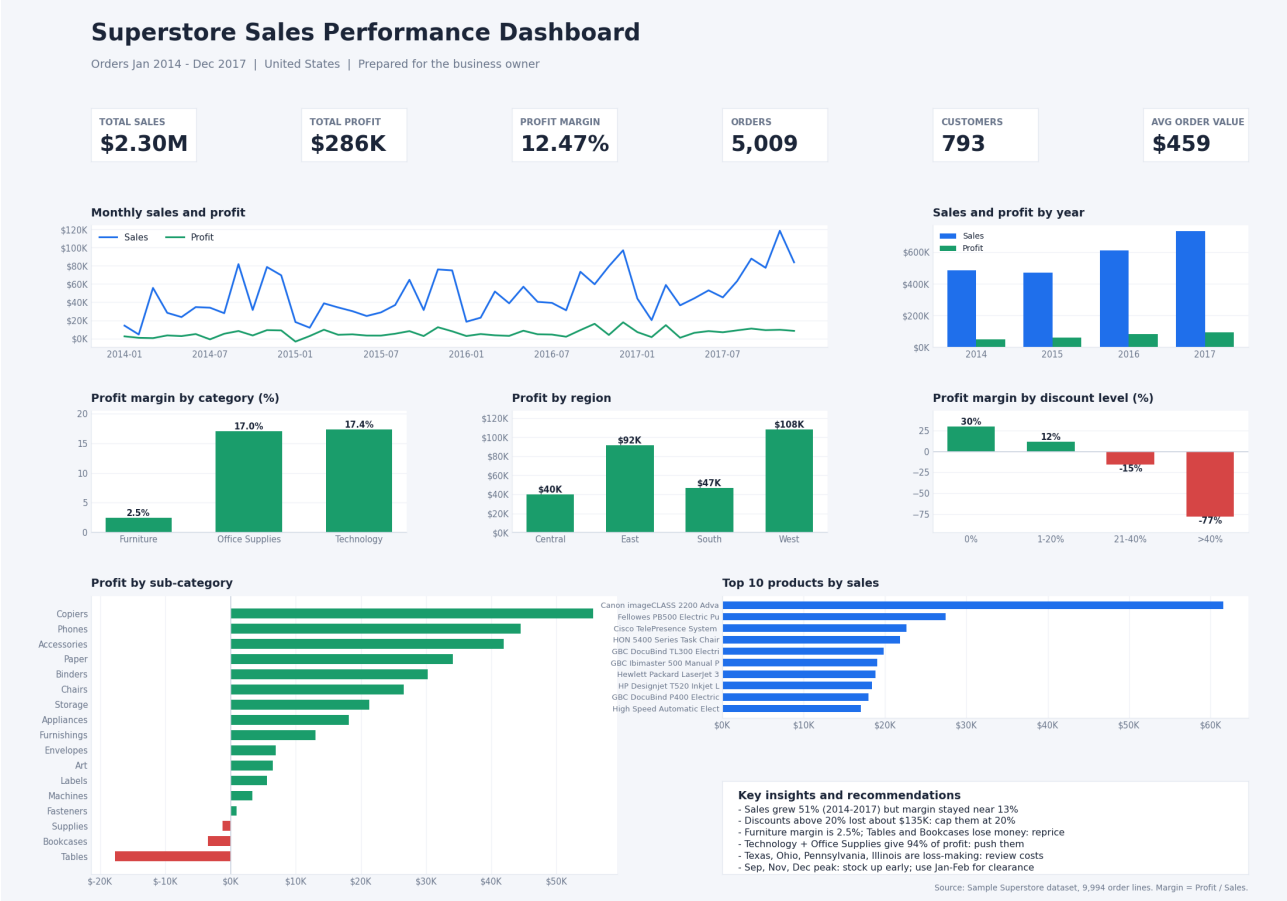
Key Findings

- Sales grew from $484K (2014) to $733K (2017), but profit margin stayed near 13%.
- Discounts above 20% are loss-making: the 21-40% band runs at -15% margin and above 40% at -77%, about $135K lost in total.
- Furniture earns only 2.5% margin. Tables (-$17.7K) and Bookcases (-$3.5K) lose money.
- Technology and Office Supplies produce about 94% of total profit at roughly 17% margin.
- West and East lead on profit. Central has the lowest margin (7.9%); Texas, Ohio, Pennsylvania and Illinois are unprofitable.
- Sales peak in September, November and December; January and February are the weakest months.

Recommendations

- Cap discounts at 20% and require approval for anything higher.
- Reprice Tables and Bookcases, or renegotiate supplier costs.
- Review pricing and shipping costs in the four loss-making states.
- Promote high-margin lines (Copiers, Accessories, Office Supplies) in marketing.
- Build stock and run campaigns before September and November; use January and February for clearance and bundles.

Dashboard



Repository Files

File	Purpose
dashboard.png / dashboard.pdf	Visual dashboard
analysis.py	Data cleaning and KPI script
make_dashboard_image.py	Generates the dashboard image
cleaned_superstore.csv	Cleaned dataset
dashboard_data.json	Aggregated data used by the charts
Sample_-_Superstore.csv	Raw data

How to Run

```
pip install pandas matplotlib
python analysis.py
python make_dashboard_image.py
```

Keep Sample_-_Superstore.csv in the same folder as the scripts.

Tools Used

Python (pandas, matplotlib), ReportLab.